Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Concepts that I learned in class that I expect to use in my program include for-loops, while-loops, conditionals, lists, dataclasses, and nested data. I want to improve my skills with Drafter and route functions while completing this project.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the easiest part of my project to be programming the system that lets the user add terms, which will utilize a list of dataclasses. I am confident in my ability to use the split function to allow the user to import large amounts of terms at once. I think the hardest part of the program will be Drafter implementation because I am not completely comfortable with using all of the Drafter functions.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I estimate that programming the multiple-choice quiz system will probably take the most time and effort. Before adding that to the project, the ability to add, remove, and view terms should be implemented first.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Getting notifications from my phone might distract me while I am working. I will minimize distractions by silencing my phone and putting it away. While working, I will need to be able to use Drafter documentation as a resource to figure out how to implement everything.

1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

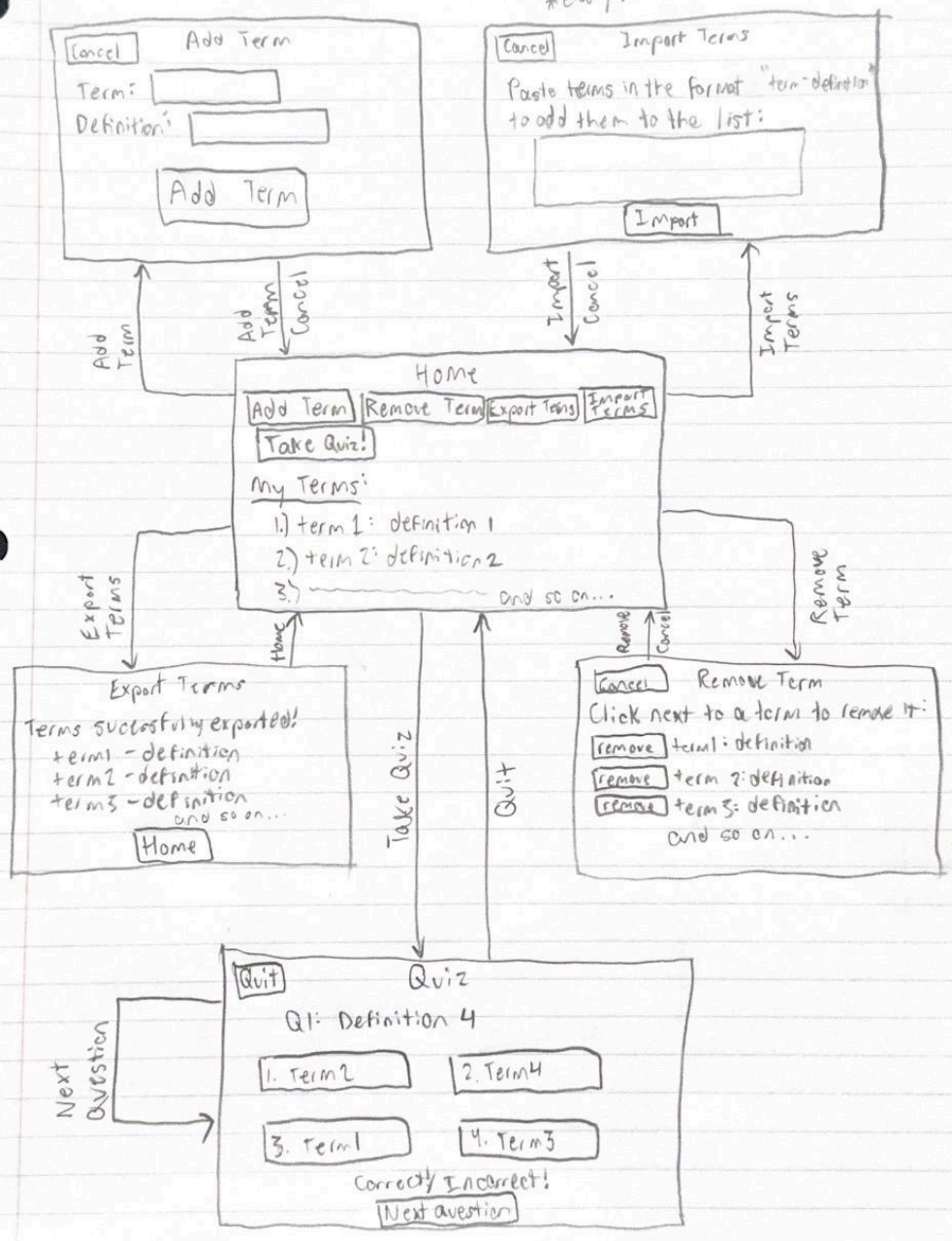
Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Class Term:
term: str
definition: str

Class State:
user_terms: list[Term]
quiz_terms: list[Term]
current_question: int
score: int

Feedback: str
answer_choices: list[str]

*every route will have state as a parameter.



\2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	My site will allow users to import vocabulary terms and definitions and take a shuffled multiple choice quiz to help them study.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	1. user_terms (list[Term]) a. Stores all terms and definitions that the user has added via the “Add Term” or “Import Terms” pages. 2. quiz_terms (list[Term]) a. Stores a shuffled version of the user’s terms for the current quiz. The list will be shuffled when a new quiz begins, and questions answered incorrectly will be appended to the end of this list so that the user has to attempt them again. 3. current_question (int) a. Stores the question number that the user is currently answering in a quiz. It will increase when the “Next Question” button is pressed. 4. score (int) a. Stores the number of questions the user has answered correctly in the current quiz. It will increment when the user chooses the correct answer to a question. 5. feedback (str) a. Stores the feedback displayed after the user answers a question. It will change to “Correct” or “Incorrect” based on the user’s answer and will reset when a new question is shown.

Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Class name: Term Attributes: ● term (str) ○ Stores the name of the vocabulary word entered by the user. It will not be modified after it is created. ● definition (str) ○ Stores the user-entered definition that goes with the term. It will not be modified after it is created.
Routes (minimum 7) The names of your route functions	1. home a. Displays the “Home” page 2. add_term a. Displays the “Add Term” page 3. import_terms a. Displays the “Import Terms” page 4. export_terms a. Displays the “Export Terms” page 5. remove_term a. Displays the “Remove Term” page 6. start_quiz a. Starts a new quiz by shuffling the user terms and resetting the score and the current question 7. quiz a. Displays the current quiz question and its four answer choices. Also processes the user’s answer and displays feedback 8. next_question a. Moves to the next question after the user has answered. If the previous answer was incorrect, it appends that question to the end of the quiz
Pages (minimum 5) A brief description of the pages in your site	1) Home a) The main page of the site that the user will return to. It shows a list of all the terms the user has added so far. This page features buttons that allow the user to add a term, import terms, export terms, remove a term, or take a

	multiple-choice quiz. 2) Add Term a) This page features two textboxes that take in user input of a term and its definition. Once the user enters their input, they can click an “Add Term” button to add this term to the list of terms and return to the home page. The user can press “Cancel” at any time to return to the home page. 3) Import Terms a) This page features one textbox that takes in terms and definitions in the form “Term - Definitions”. The user can add multiple terms at once, with each term separated by a new line. The main purpose of this page is to allow the user to paste a large amount of terms at once. Once the user enters into the textbox, they can press an “Import” button to officially add all of the terms, as long as they are in the right format. The user will be returned home after importing. The user can press “Cancel” at any time to return to the home page. 4) Export Terms a) This page takes all of the terms that you have inputted and displays them in the form “Term - Definition”, with each term separated by a new line. The purpose of this page is to allow the user to save their inputted terms somewhere else on their computer so that they can import them back in the next time that they use the program. There is a “Home” button at the top so that the user can return to the home page at any time. 5) Remove Term a) This page shows you all of the terms that you have added so far and has a button next to each of them to remove that term. After removing a term, the user returns to the home page. The user can also press “Cancel” to go back to the home page.

	6) Quiz a) This page has text at the top that displays a random definition out of all of the terms that the user has added. Below that, there are four buttons that each display a random term out of all of the terms the user has added. One of these terms will always be the correct term that matches to the given definition. If the user presses the button that matches to the given definition, text will appear that says “Correct” and a button will appear that allows the user to go on to the next question. If the user presses one of the three incorrect buttons, text will appear that says “Incorrect,” it will tell the user the correct answer, and a button will appear that allows the user to go on to the next question. The question that was guessed incorrectly will be added to the end of the quiz so that the user has to attempt it again. When the user presses the button to go to the next question, the text at the top will change to a different definition and the answer choices will also change, resetting the cycle. At any point on this page, the user can press a “Quit” button in the top left to exit back to Home.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	1. TextBox a. Used on the “Add Term” page to allow the user to enter terms and definitions. 2. TextArea a. Used on the “Import Terms” page to allow the user to paste multiple terms and definitions at once, each separated by a new line. 3. Button a. Used on every page for actions such as returning to the home page, adding terms, removing terms, selecting quiz choices, and moving to the next question.

If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	1. Check whether user_terms contains at least four terms before starting a quiz a. If the user has less than four terms, a message will be shown saying that they need to add at least four terms before taking a quiz. Otherwise, the quiz will start. 2. Check whether the user's imputed term and definition on the "Add Term" page is valid a. If the term or definition are empty, an error message will be shown and no term will be added. Otherwise, a new instance of Term is created and added to the list. 3. Check if imported terms are in the required format a. If an import does not contain the separator " - ", it will not be added to the list. Otherwise, the terms will be added to the list. 4. Check whether the user's chosen answer is equal to the correct answer a. If the chosen answer is correct, the website shows "Correct" and increases the score by 1. Otherwise, it will show "Incorrect" and reveal the correct answer. 5. Check whether the user has reached the end of the quiz a. If there are no more questions remaining, the website will show the final score and an option to return to the Home page. Otherwise, the next question is shown.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	for term in state.user_terms ● Iterates through every instance of Term in the user's term list. This loop will be used on the "Home" page to display all of the terms and definitions. Similarly, it can also be used on the "Export Terms" page to combine all vocab terms into one string, and on the "Remove Term" page to show each term with a button to remove that term.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I chose not to use an LLM to assist me with this assignment.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.